

2023 Higher Education Press Cup — China Undergraduate Mathematical Contest in Modeling

Problem A: Optimal Design of a Heliostat Field

(Please first read the “Formatting Specification for CUMCM Papers.”)

Translator's note: Unofficial English translation of the official Chinese problem statement, prepared for study. Symbols, constants, and formulas follow the original exactly; figures are referenced by caption as in the source. In case of any discrepancy, the official Chinese text prevails.

Background

Building a new-type power system with new energy as the mainstay is an important measure for China to achieve its “carbon-peak” and “carbon-neutrality” goals. Tower solar thermal (concentrating solar) power generation is a low-carbon, environmentally friendly clean-energy technology [1].

A heliostat is the basic component by which a tower CSP station (hereafter “tower station”) collects solar energy. Its base consists of a vertical (azimuth) shaft and a horizontal (elevation) shaft, with the flat mirror mounted on the horizontal shaft. The vertical shaft’s axis is perpendicular to the ground and controls the mirror’s azimuth angle; the horizontal shaft’s axis is parallel to the ground and controls the mirror’s pitch (elevation) angle (the heliostat and base are shown schematically in Figure 1). The height above ground of the intersection of the two shafts — which is also the heliostat center — is called the heliostat’s install height. A tower station uses a large array of heliostats, called a heliostat field. The heliostats reflect and concentrate sunlight onto a receiver mounted at the top of the absorption tower in the field, heating the heat-transfer medium inside and storing the solar energy as heat, which is then converted from heat into electricity through heat exchange. Sunlight is not parallel rays but a cone-shaped beam with a certain cone angle, so the rays reflected from any point of a heliostat also form a cone-shaped beam [2]. During operation, the control system adjusts each heliostat’s normal in real time according to the sun’s position, so that the ray emitted from the center of the sun, after reflecting at the heliostat center, points to the center of the receiver. The height of the receiver center above the ground is called the absorption-tower height.

[Figure 1. Schematic of a heliostat and its base — illustration in the original.]

A circular heliostat field is now planned within a circular region centered at 98.5° E longitude, 39.4° N latitude, altitude 3000 m, with radius 350 m (Figure 2). A coordinate system — the field coordinate system — is established with the center of the circular region as the origin, due east as the $+x$ axis, due north as the $+y$ axis, and vertically upward as the $+z$ axis.

The planned absorption-tower height is 80 m; the receiver is a cylindrical, external-surface light-receiving receiver 8 m tall and 7 m in diameter. No heliostats are installed within 100 m of the tower, leaving open ground to build a plant housing the power-generation, energy-storage, control, and other equipment. Each heliostat is a flat rectangle whose top and bottom edges are always parallel to the ground; the distance between these edges is the mirror height, and the distance between the left and right edges is the mirror width (normally the mirror width is not less than the mirror height). The mirror edge length is between 2 m and 8 m, and the install height is between 2 m and 6 m; the install height must ensure the mirror does not touch the ground while rotating about the horizontal shaft. Because maintenance and cleaning vehicles must pass, the distance between the base centers of adjacent heliostats is required to exceed the mirror width by more than 5 m.

To simplify the calculation, all “annual-average” indicators in this problem are evaluated at the local times 9:00, 10:30, 12:00, 13:30, and 15:00 on the 21st of each month.

[Figure 2. Schematic of a circular heliostat field — illustration in the original.]

Problems

Please build models to solve the following problems.

Problem 1

If the absorption tower is built at the center of the circular field, all heliostats are $6\text{ m} \times 6\text{ m}$ with install height 4 m , and the positions of all heliostat centers are given (hereafter “heliostat positions”; see the attachment for the data), compute the field’s annual average optical efficiency, annual average output thermal power, and annual average output thermal power per unit mirror area (the definitions of optical efficiency and output thermal power are given in the Appendix). Fill the results into tables in the formats of Table 1 and Table 2.

Problem 2

By design requirement, the field’s rated annual average output thermal power (hereafter “rated power”) is 60 MW . If all heliostats have the same size and install height, design the following field parameters: the absorption-tower position coordinates, the heliostat size, the install height, the number of heliostats, and the heliostat positions — so that, while meeting the rated power, the annual average output thermal power per unit mirror area is as large as possible. Fill the results into tables in the formats of Tables 1, 2, and 3, and save the absorption-tower position, heliostat size, install height, and position coordinates to the file result2.xlsx in the template’s prescribed format.

Problem 3

If the heliostat sizes may differ and the install heights may also differ, with the rated power the same as in Problem 2, redesign all field parameters so that, while meeting the rated power, the annual average output thermal power per unit mirror area is as large as possible. Fill the results into tables in the formats of Tables 1, 2, and 3, and save the absorption-tower position, each heliostat’s size, install height, and position coordinates to the file result3.xlsx in the template’s prescribed format.

Result Tables

Table 1. Problem X — average optical efficiency and output power on the 21st of each month.

Date	Avg optical eff.	Avg cosine eff.	Avg shadow-blocking eff.	Avg truncation eff.	Per-unit-area avg output (kW/m^2)
Jan 21					
Feb 21					
Mar 21					
Apr 21					
May 21					
Jun 21					
Jul 21					
Aug 21					
Sep 21					
Oct 21					

Date	Avg optical eff.	Avg cosine eff.	Avg shadow-blocking eff.	Avg truncation eff.	Per-unit-area avg output (kW/m ²)
Nov 21					
Dec 21					

Table 2. Problem X — annual average optical efficiency and output power.

Annual avg optical eff.	Annual avg cosine eff.	Annual avg shadow-blocking eff.	Annual avg truncation eff.	Annual avg output (MW)	Per-unit-area annual avg output (kW/m ²)

Table 3. Problem X — design parameters.

Absorption-tower position	Heliostat size (W × H)	Install height (m)	Total number (mirrors)	Total mirror area (m ²)

Note: When filling Problem 3's results into Table 3, the "Heliostat size" and "Install height" columns may be left blank.

Appendix — Relevant Formulas

1. Solar angles

Solar altitude angle α_s [3]:

$$\sin \alpha_s = \cos \delta \cos \varphi \cos \omega + \sin \delta \sin \varphi$$

Solar azimuth angle γ_s [4]:

$$\cos \gamma_s = (\sin \delta - \sin \alpha_s \sin \varphi) / (\cos \alpha_s \cos \varphi)$$

where φ is the local latitude (north positive) and ω is the solar hour angle:

$$\omega = (\pi / 12)(ST - 12)$$

where ST is the local time and δ is the solar declination [5]:

$$\sin \delta = \sin(2\pi D / 365) \cdot \sin(2\pi \cdot 23.45 / 360)$$

where D is the number of days counted from the spring equinox as day 0 (for example, if the spring equinox is March 21, then April 1 corresponds to D = 11).

2. Direct normal irradiance (DNI)

DNI (kW/m²) is the solar radiant energy received per unit area per unit time on a plane perpendicular to the sunlight at the Earth's surface, approximated by [6]:

$$DNI = G_o \cdot [a + b \cdot \exp(-c / \sin \alpha_s)]$$

$$a = 0.4237 - 0.00821(6 - H)^2$$

$$b = 0.5055 + 0.00595(6.5 - H)^2$$

$$c = 0.2711 + 0.01858(2.5 - H)^2$$

where G_o is the solar constant, taken as 1.366 kW/m², and H is the altitude (unit: km).

3. Field output thermal power

$$E_{field} = DNI \cdot \sum(i=1..N) A_i \cdot \eta_i$$

where DNI is the direct normal irradiance; N is the total number of heliostats (unit: mirrors); A_i is the light-collecting (aperture) area of the i-th heliostat (m^2); and η_i is the optical efficiency of the i-th mirror.

4. Heliostat optical efficiency

$$\eta = \eta_{sb} \cdot \eta_{cos} \cdot \eta_{at} \cdot \eta_{trunc} \cdot \eta_{ref}$$

where:

- Shadow-blocking efficiency: $\eta_{sb} = 1 - (\text{shadow-blocking loss})$
- Cosine efficiency: $\eta_{cos} = 1 - (\text{cosine loss})$
- Atmospheric transmittance:

$$\eta_{at} = 0.99321 - 0.0001176 \cdot d_{HR} + 1.97 \times 10^{-8} \cdot d_{HR}^2 \quad (d_{HR} \leq 1000) \quad [7]$$

- Receiver truncation efficiency:

$$\eta_{trunc} = (\text{energy received by the receiver}) / (\text{total mirror-reflected energy} - \text{shadow-blocking loss energy})$$

- Mirror reflectivity η_{ref} may be taken as a constant, e.g., 0.92.

Here d_{HR} is the distance from the mirror center to the receiver center (unit: m).

References

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- [5] Cai, Z. Solar Shadow Positioning [J]. Mathematical Modeling and Its Applications, 2015, 4(4): 25–33.
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